Universality in Quantum Mechanics:
A Claim That Cannot Be Tested, and One That Can

Claes Johnson[1]

[1]KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,
claesjohnson@gmail.com

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The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble. — P. A. M. Dirac, 1929 [6]

Abstract

Standard quantum mechanics is held universal for chemistry: one parameter-free equation, the nuclear charges its only input. We ask what grounds that claim. It is not one equation — it is Schrödinger plus spin, plus antisymmetry, plus the spin–statistics connection binding them, three components none of them derived. RealQM, a model of non-overlapping unit electron charge densities in three space dimensions, is equally parameter-free and adds one, which supplies both the exclusion and the shell capacities 2 and 8. Within chemistry, the domain at issue here, spin proves unnecessary for both of the things it is chiefly invoked to do.

More decisively, the standard claim cannot be tested. In $3N$ dimensions the cost grows exponentially, so past a few tens of electrons the equation is uncomputable and what is computed instead is a parameterised reduction. Its agreement with measurement does not transfer to the equation behind it, and the approximation hierarchy cannot carry it upward, that hierarchy's exact limit being the uncomputable object itself. Universality is verified at one to three electrons and assumed beyond, shielded by intractability.

RealQM computes at linear cost, so its universality is empirical: atoms to radon, the row capacities of the periodic table, H_2 at 94% of the measured dissociation energy, water's dipole and angle at 99% and 97%, hydrogen-bond geometries matching measurement, and polypeptides of 153 residues carried with their electrons on a laptop. Its failures are attributable too. The asymmetry is dimensional, not technical, and we close with three tests that would decide.

1 What a universality claim asserts

The claim examined here is not a straw man, and it is worth establishing that before examining it. Dirac stated it in 1929 in the sentence above, and stated it in its strongest form: the laws sufficient for *the whole of chemistry* are completely known. What makes the sentence remarkable is that it asserts the uncomputability in the same breath — the equations are "much too complicated to be soluble" — and treats that as a difficulty of application rather than as bearing on the claim

itself. The programme that followed has largely kept both halves. The term *ab initio* is itself an assertion of derivation from first principles, and the question whether chemistry reduces to the Schrödinger equation remains live rather than settled [23], with Löwdin's challenge over the $(n+\ell)$ rule presupposing that it does [21, 22].

A careful practitioner will qualify all this, and the qualification should be granted rather than argued with: what is claimed is that the equation is exact *in principle*, not that any particular computation is universal, and the limitations of each method are generally well understood by those using it. That qualification is precisely the subject of this article. An in-principle claim is worth whatever its principle can be made to license, and the question we press is what an in-principle universality licenses when the principle can never be applied to any system for which the claim would be interesting.

A model is universal over a domain of phenomena if three conditions hold.

1. **Species-blindness.** The model makes no reference to the kind of system it is applied to. It does not mention atoms, bonds, metals or proteins; one writes the same thing for helium and for a protein, differing only in Z and the geometry.

2. **Closure under composition.** Joining two systems yields the same model for the union. There is no boundary in the domain at which one must change theories.

3. **Accountability.** The model's predictions can be compared with measurement across the domain claimed, and the comparison can go against it.

The first two are formal and easily checked; both models satisfy them completely. The third is not formal at all, and it is the one that does the work. A model satisfying only the first two is a *framework*, not a universal theory, because nothing it says about most of its domain can be held to anything.

2 The components of each model

Two models of electronic matter are on the table. Both take the nuclear charge Z as their only supplied number, both use the same law — kinetic energy plus Coulomb interaction — and neither fits anything. They differ in what must be *added* to that law before it describes matter, and it is worth laying the additions side by side before any question of accuracy arises.

The domain in question throughout this article is **chemistry**: atoms, molecules, materials and biomolecules, their energies and their geometries. Magnetic phenomena — the Zeeman effect, Stern–Gerlach, the electron's magnetic moment — lie outside it and are not treated here. That restriction is not a convenience: within chemistry all three of the standard formalism's additions carry weight, spin supplying the capacity two, antisymmetry the shell structure, and the spin–statistics connection the link between them, so the comparison below is made where the components are all doing work.

	standard quantum mechanics	RealQM
state	$\Psi(x_1, \ldots, x_N)$ on $\mathbb{R}^{3N}$ [1], complex, antisymmetric	N charge densities on $\mathbb{R}^3$, real, non-overlapping
supplied number	Z	Z
law	kinetic + Coulomb	kinetic + Coulomb
added component 1	**spin**: a two-valued internal degree of freedom, $s = \frac{1}{2}$ [2, 8]	**non-overlap**: densities occupy disjoint domains with free interfaces
added component 2	**antisymmetry** of Ψ, from which Pauli exclusion [3, 4] follows	—
added component 3	**spin–statistics** [5]: the rule that half-integer spin takes the antisymmetric representation	—
configuration	multiplicity and orbital occupations — an output of full configuration interaction, which *cannot be run*	the build-up is computable within the model and is carried out to carbon; beyond it the filling is imported, the obstacle being the present outward-only search

Three additions against one. Several remarks are needed to make that count honest in both directions.

Pauli is not a fourth component. Exclusion is a consequence of antisymmetry, not an independent postulate alongside it. What *is* independent is the spin–statistics connection: antisymmetry by itself does not say which particles it applies to. In nonrelativistic quantum mechanics that connection is a postulate; it becomes a theorem only after one adopts relativistic quantum field theory [5], which is a far larger commitment than the equation one began with.

None of the three is derived. Each was added because observation demanded it. The parameter-free Schrödinger equation on its own is content with symmetric solutions and predicts the wrong periodic table; it is the additions that produce chemistry.

Does spin enter as a parameter? This deserves a direct answer, and it has two halves. The *value* does not: $s = \frac{1}{2}$ is a fixed postulated constant, fitted to nothing, and $g = 2$ follows once the Dirac equation is in hand. In that sense spin is structure rather than a parameter. But the *assignment* does, in every calculation actually carried out: one specifies the multiplicity and the orbital occupations — that this system is a singlet, that one a triplet — and the computation does not determine them. In principle a full configuration interaction in a complete basis would return the spin state as an output rather than take it as an input, so the assignment is not a parameter of the theory. That defence, however, rests on exactly the calculation which cannot be performed, which is the subject of Section 4.

Both import a configuration, and the imports are not alike. RealQM derives the shell capacities at every Z and performs the build-up explicitly through lithium, beryllium, boron and carbon; beyond carbon the filling is imported. So a configuration is supplied in each case, and it would be easy to call that a draw. It is not one, and the difference is the subject of this article.

The standard formalism's multiplicity is an output of a full configuration interaction which *cannot be run*, now or by any method expected to exist. The limitation is permanent. RealQM's build-up, by contrast, is an ordinary computation within the model — it is what produces the configurations through carbon — and what stops it beyond carbon is that the procedure presently used adds charge outward, so it cannot place a charge inside a shell already occupied. That is a restriction on the search, not on the model, and it is removed by writing a different search rather than by waiting for an intractable problem to become tractable.

So the honest statement is that the Aufbau build-up is computable in RealQM in principle and carried out to carbon in practice. One of the two limitations has a known route out of it; the other does not.

What the single postulate buys. Non-overlap does the work of components 2 and 3 together: electrons exclude one another because they cannot occupy the same region, so no antisymmetry need be imposed and no statistics rule is needed to say whom it applies to. It also supplies the shell capacities, which the standard formalism obtains from spin multiplied by orbital degeneracy — the antipodal pair gives 2, and the octet $8 = 4{+}4$ follows from the same packing rule, two dual tetrahedra — the decomposition of Linnett's double quartet [19], reached here from Coulomb interaction and non-overlap rather than from spin — since a lone tetrahedron has no inversion centre and cannot close.

Is spin necessary? The question has no single answer because it is three questions.

- **For exclusion: no.** Territory does the work, and this is not a conjecture — it is what every atomic result in Section 5 is computed from.
- **For the capacity two: no.** The antipodal pair gives it geometrically, and the same rule gives eight. Neither borrows the spin degeneracy.
- **For the two-valued magnetic response: outside this article.** A scalar charge density gives one undeflected Stern–Gerlach spot, not two with no middle, so something further is required there. Whether that something must be spin as the standard formalism has it is a real question and not one we settle; it belongs to the magnetic phenomena excluded above.

Within chemistry, then, spin is not necessary for either of the things it is chiefly invoked to do. And it should be said that neither model *explains* spin: the standard formalism posits it. Its presence there cannot be defended as something nature requires and an alternative merely lacks — it is an addition, like the other two, made because observation demanded it. What is distinctive is that the standard formalism needs it twice over: once as spin, and again as the statistics rule which spin is then made to govern.

A universal model should cover more with less. On that measure the comparison above favours the one-postulate model before any question of accuracy is reached, and the rest of this article asks whether either model's universality can be tested at all.

3 How many parameters, and where from

Being wholly free of parameters is an ideal rather than a practice. Every model that reaches a large system does so through a reduction, and reductions carry constants. So the useful questions are not whether a model has parameters but *how many, to reach what scale*, and — decisively — *where they come from.*

That last question admits two answers which are not epistemically alike.

- **Upscaled**: the constant is computed from the model itself at a finer scale. The coarse model then has a parameter, but it was derived rather than fitted, and a prediction made with it remains a prediction. This is ordinary multiscale practice and it is legitimate.
- **Fitted to observation**: the constant is adjusted so that the model reproduces measured data. If the data fitted is the data later predicted, the agreement is circular and tests nothing. If it is a *different* observable, the agreement is a weaker but real test: one quantity is spent to buy another.

Both models use both kinds. Setting them out by scale is more informative than any global claim about parameter-freedom:

scale	RealQM	standard quantum mechanics
atoms	**none**; Z alone, all-electron on the bare kernel $-Z/r$	**none fitted to experiment** — but basis-set exponents are optimised by minimising *atomic* energies, so an atomic energy computed in that basis is not an independent prediction
small molecules	**none**, where the atoms are treated all-electron	**none fitted to experiment** in the CCSD(T) near-basis-set-limit window; the basis is upscaled from atomic calculations, though its form and truncation are chosen and the basis-set limit itself is reached by a fitted extrapolation
larger molecules, materials	**none in the model**, which is defined all-electron on the bare kernel. In practice a core radius r_c avoids resolving the $1s$ on a molecular mesh — a numerical device, like the mesh itself, and not a component. But it is not demonstrably benign: see below	a choice of exchange–correlation functional, several of them **fitted to benchmark sets** of the properties later computed
dispersion	absent from the model	an added term **fitted to reference data**
biomolecules	r_c, plus the **secondary structure** (which residues are helical, driving a hydrogen-bond bias) and a **surface-area term** standing in for the hydrophobic effect. Tertiary packing is *not* biased	not reachable; classical force fields reach it, **wholly fitted** to experiment and to quantum calculation

Three things follow, and the second is not favourable to the argument of this article.

r_c is a numerical device, not a component — but it is not a clean one. The core radius deserves separating from the other entries, because it is a different kind of object. The model needs no r_c: every atom in Section 5 is computed all-electron on the bare kernel $-Z/r$, hydrogen to krypton, which settles that r_c is not part of the formulation. What it does is numerical. A molecule needs a box spanning the whole molecule, while the innermost shell contracts as $r_{1s} \approx 4.3/Z$, and a single uniform mesh cannot do both; replacing the resolved core by a reduced kernel and an exclusion radius buys the molecular calculation at the cost of a length. In that respect r_c belongs with the mesh spacing rather than with a fitted constant, and a model carrying only numerical parameters carries none that matter for the present argument.

That is the favourable reading and we should not take it without the qualification. A benign numerical parameter does not change answers qualitatively, and this one does. On the hydroxyl fragment, with the oxygen valence split and the boundaries free, $r_c = 0$ gives a dipole of 1.98 D

with the oxygen negative, against a measured 1.66 D; at $r_c = 0.5$ the same configuration gives 0.34 D *with the polarity inverted*. A mesh spacing that reversed the sign of a dipole would be called a broken discretisation, and the same standard applies here.

It is worth being quantitative about what r_c actually buys, because the usual phrase — that all-electron molecules are too expensive — turns out to name the wrong cost. For a protein built from hydrogen, carbon, nitrogen and oxygen the *electron count* barely moves: per residue the reduced description carries about 35 electrons against 59 all-electron, and for a 153-residue chain 5340 against 9042, a factor 1.7. That is not what makes the calculation hard. The cost is the *mesh*. The run above uses 200^3 cells on a 100 bohr box, so $h = 0.5$ bohr, while resolving an oxygen $1s$ — $r_{1s} \approx 4.3/Z \approx 0.54$ bohr, spanned by some ten cells — requires $h \approx 0.054$. That is a factor 9.3 in each direction and therefore some 800 in cells, a 200^3 grid against roughly 1860^3; with cost going as electrons times mesh points the product is of order 10^3.

But that factor is only needed within about a bohr of each heavy nucleus. With roughly twelve hundred carbon, nitrogen and oxygen atoms, core-refined patches of radius one bohr at $h = 0.054$ come to some 6×10^7 cells in total, against 6.4×10^9 for refining everywhere, which places an all-electron calculation of the same chain at something like fifteen times the present cost rather than a thousand. So r_c is not required by the physics, as the all-electron atomic results show, and it is not required by the cost either. It is required by the use of a *uniform* grid. On a mesh with local refinement it would disappear rather than be calibrated — which, given that it reverses a dipole, is the better outcome. Two details make the present situation less comfortable still: at $h = 0.5$ the oxygen core radius of 0.8 bohr spans one and a half cells, so the exclusion sphere is represented by a cell or two and the polarity inversion may itself be a resolution artefact; and the $r_c \to 0$ limit has never been taken for a molecule, that being precisely the calculation r_c exists to avoid.

So the honest position is that RealQM has no model parameters at any scale, and one numerical device which is an artefact of uniform meshing and whose convergence has not been demonstrated where it is used.

What "nothing fitted" does and does not mean for the standard formalism. It means nothing fitted to experiment, and that is worth less than it sounds. Three of the four qualifying remarks are mechanical: the basis form is a choice rather than a derivation, Gaussians with a chosen primitive count and contraction pattern; the basis-set limit, which is the rung supposed to be free of constants, is approached through an extrapolation formula carrying fitted ones; and effective core potentials for the heavier elements are fitted to reference data outright. The fourth is more serious. Basis-set exponents are determined by minimising *atomic* energies, so an atomic energy computed in such a basis is not an independent prediction — the variational principle largely guarantees a good number from a basis tuned to that very system. The circularity is milder for molecules, which is why basis-set superposition error exists as a named effect; but it means the atomic scale, where the standard formalism looks cleanest, is in fact its least independent test.

The clean regimes coincide, and neither model is tested outside them by its own successes. For atoms and small molecules both models can be run with no constant fitted to the predicted quantity, and both should be believed there, with the reservation just entered. RealQM's advantage in that band is coverage rather than kind: it runs all-electron across the table and into the fourth row at a few per cent, where the standard formalism's clean window

is a few tens of atoms at a far higher precision. That is a genuine trade and not a defeat either way.

RealQM's parameters become circular at exactly the scale where its scope claim is most striking. The core radius is numerical, as just discussed. At the protein scale the secondary structure is supplied — which segments are helical, which then drives a hydrogen-bond bias — together with a two-parameter surface term standing in for the hydrophobic effect. The tertiary packing is not biased and is computed, so what those runs cannot be offered as is evidence that the model predicts secondary structure. The honest reading is that RealQM's parameter count rises from zero to one to three as the scale grows, and that the last step crosses from upscaled to fitted. Any claim resting on the largest systems must be read in that light.

But the crossing happens earlier for the standard formalism, and for everything. Outside the CCSD(T) window — which is to say for every system in materials science, catalysis and biology — the computed object is a functional-bearing reduction, and the functionals are in substantial part fitted to the properties they are then used to compute. The dispersion term is fitted outright. So on the parameter question the comparison is not parameter-free against parameterised; it is *one parameter sourced from a different observable, out to materials and solids*, against *a fitted functional from small molecules onward*. Where each model stops being testable by its own output is the real content of the comparison, and RealQM's crossing comes later.

4 Why the universality of standard quantum mechanics cannot be tested

4.1 The equation is uncomputable where the claim is interesting

The state is a function on $\mathbb{R}^{3N}$, and representing it costs exponentially in N: full configuration interaction in a basis of M functions scales as $\binom{M}{N}^2$ in the worst case, and the exact solution is in practice out of reach past roughly twenty electrons. This is not a matter of waiting for better machines. The ground-state problem for a local Hamiltonian is QMA-complete [10, 9], so no efficient general algorithm is believed to exist even on a quantum computer; the barrier has the character of a theorem rather than of an engineering limit.

The systems for which universality is an interesting claim — a catalyst, a protein, a crystal, a cell — carry 10^3 to 10^{23} electrons. For every one of them the model cannot be solved.

4.2 What is computed instead is not the model

Since the equation cannot be solved, it is not the equation that is compared with experiment. What is compared is a reduction, and every reduction in use carries parameters. Gaussian basis sets are built by optimising exponents and contraction coefficients atom by atom, and the complete-basis-set limit is reached by extrapolation through a fitted functional form. Coupled-cluster theory [13] truncates the excitation operator, and CCSD(T)'s perturbative triples correction [14] is not variational and carries no bound — its celebrated kcal/mol accuracy is an empirical regularity over well-behaved closed-shell molecules which degrades without warning for multireference systems. Density functional theory [11, 12], which performs essentially all practical calculation on large systems, selects an exchange–correlation functional from a zoo in

which B3LYP, PBE, M06 and ωB97X are different theories with different failures, several fitted to benchmark sets outright.

We should be fair about the strongest case. There is a window — small closed-shell molecules of up to a few tens of atoms — in which CCSD(T) near the basis-set limit is nearly parameter-free and agrees with measurement to about a kcal/mol across a wide range of chemistry. That is a real achievement and we do not dispute it. But it is a window, and outside it, which is to say for almost everything, the computed object is a functional-bearing reduction.

4.3 The hierarchy cannot transfer verification upward

It is often said that this does not matter, because the approximations form a convergent hierarchy — Hartree–Fock, MP2, CCSD, CCSD(T), full CI — whose limit is the exact solution, so the hierarchy connects what is computed to what is claimed.

It does not, for a structural reason. The hierarchy has two ends and both are unavailable. Its exact limit is the uncomputable object itself, so one can never stand at the top and check. Its lower rungs, which are what gets computed, each carry parameters, so agreement there is agreement of a parameterised procedure. There is no rung which is both reachable and parameter-free, and consequently no step at which empirical success attaches to the parameter-free equation rather than to the reduction. The existence of a convergent sequence establishes that the reductions approach something; it does not establish that the something is nature, since that identification is exactly what was to be tested.

4.4 The practice this produces

The consequence is visible in how discrepancies are handled. When density functional theory fails on dispersion — and semilocal and hybrid functionals do fail on it, systematically — the response is to add an empirical correction, a fitted term [15], and the failure is charged to the functional. When a coupled-cluster result is wrong for a transition-metal complex it is charged to the truncation or the basis. In no case is it charged to the Schrödinger equation, and this is not dogmatism: it is forced, because no computation of the equation exists against which the equation could be found wanting.

A claim that no measurement can contradict is not thereby strong; it has been removed from the class of claims that measurement bears on. Over the great majority of its asserted domain that is the position of the universality of standard quantum mechanics — verified superbly at one to three electrons, assumed beyond.

5 Why the universality of RealQM can be tested

RealQM is in the opposite position on the one point that matters, for one reason: it is computable where the claim is interesting. The state is N densities on $\mathbb{R}^3$, so the cost is the electron number times the number of mesh points — linear in N, not exponential. A molecule, a cluster, a crystal, a protein is the same computation with more kernels and more domains. The model itself, with Z as its only input and no reduction interposed, can be run on the systems in question and the output compared with measurement.

That makes universality an empirical question in the sense of Section 1(3). What follows is the record. It is organised by scale, because the point is not any single number but that the same four-line scheme, unchanged, reaches every scale at which chemistry happens.

5.1 Atoms

Hydrogen is an identity, not a test. For one electron there is no exclusion and no partition, and RealQM *is* the Schrödinger equation: both give $E = -0.5$ Ha and $\alpha = 4.5\, a_0^3$. We record it because it locates where the two models can first differ — at two electrons, where the exclusion postulate enters.

Helium. The first real test, with the answer known exactly.

	E (Ha)	error
exact (nonrelativistic) [16]	-2.9037	—
RealQM, two domains, front-tracked plane	-2.91	0.24%
Hartree–Fock	-2.8617	1.4%

The two-domain result is closer to exact than Hartree–Fock, recovering about two thirds of the correlation energy without any correlated wave function. Territorial exclusion, replacing antisymmetry, is not merely tolerable on helium's energy; it is better than the mean-field theory it stands beside. Lithium and the lithium cation follow the same pattern against their exact values of -7.478 and -7.280 Ha.

The table. Computed all-electron on the bare kernel $-Z/r$, with no core radius, no fitted length and nothing adjusted per element: through argon the mean relative error is 1.6% with maximum 4.5%, and 13 of 17 atoms fall within 2.5%. Reference energies throughout are the near-exact nonrelativistic values of Chakravorty *et al.* [17]. Extended to the fourth row on a mesh chosen by Z, the mean is 3.3%, every entry under-bound in the direction the spherical reduction predicts. Energies are available through to radon.

The periodic table's capacities. Two filled packings appear and no third: the antipodal pair, 2, and the octet, $8 = 4{+}4$, two dual tetrahedra. A lone tetrahedron has no inversion centre and so cannot close a shell; its dual completes the cube, which is centrosymmetric and admits no further dual. Row capacities are sums of these two alone,

$$2, \qquad 8, \qquad 18 = 2{+}8{+}8, \qquad 32 = 8{+}8{+}8{+}8,$$

and a period ends when its capacity is exhausted, giving the closures $2, 10, 18, 36, 54, 86, 118$. The capacities obey a parity rule — remainder modulo 8 is only ever 0 or 2, never 4 or 6, so each is a whole number of octets carrying at most one pair, the pair appearing exactly when the shell index is odd — and admit a closed form, $\mathrm{cap}(n) = 8\lfloor n/2 \rfloor \lceil n/2 \rceil + 2(n \bmod 2)$, which equals $2n^2$ for every n without ever invoking it. The account coincides with the hydrogenic degeneracy rather than borrowing it.

Spectra and valence quantities. Excited-state spectra are computed for the alkalis, the alkaline earths and helium, including the helium ground-state-to-orthohelium excitation, and ionization energies for lithium and fluorine with their ions. Electronegativity comes out natively: Allen's configuration energy [20] reproduces the ordering across period 2 exactly.

5.2 Molecules

The covalent bond. H_2 on a 200^3 grid, with the binding energy taken self-referentially from the same solver as $E_{\text{bind}} = E(R_e) - E(R{=}6)$:

quantity	RealQM	reference
E at $R = 2.0\,a_0$	-1.1923 Ha	-1.1745 Ha (Kolos–Wolniewicz [18])
E at $R = 6\,a_0$	-1.0379 Ha	-1.000 Ha ($2\times$H)
E_{bind}	-0.1543 Ha (-4.20 eV)	-0.1646 Ha (measured D_0)

94% of the measured dissociation energy, from a model with no basis set and no antisymmetry. The curve dissociates correctly to two hydrogen atoms as R is increased, so the bond is formed and broken by the same scheme. Worth noting on what is being compared: of the H_2 curve only R_e and D_0 are measured quantities — the energy at arbitrary R is itself computed. The textbook six-decimal agreement for H_2 belongs to a computation, not to a measurement.

Water. The dipole moment and the bond angle are the two quantities that test whether the charge is in the right place. With the oxygen valence split into angular sectors, one toward each hydrogen and one pointing away:

quantity	RealQM	experiment	ratio
dipole moment μ	1.836 D	1.85 D	99%
H–O–H angle	101.6°	104.5°	97%
equilibrium O–H bond	$1.814\,a_0$	$1.814\,a_0$	exact

The lone pair is not put in by hand; the sector pointing away from both hydrogens is where the energy puts it. The equilibrium bond length comes out on the measured value.

5.3 Condensed matter and reactions

The hydrogen bond is where molecular matter becomes material, and it is geometric:

system	RealQM	reference
water dimer	H$\cdots$O = 1.87, O–O = 2.95 Å	matches measurement
formamide dimer	N–H = 1.01, H$\cdots$O = 2.04, N$\cdots$O = 2.97 Å	matches measurement
ice I_c, 216 molecules	O–O = 2.73 Å	2.76 Å

Beyond hydrogen bonding, the same scheme gives metallic bonding — a 100-atom lithium lattice, perturbed, restores itself — ionic crystals and their melting in sodium chloride, conduction barriers for distinct arrangements of charge, and chemical reaction paths: H + H_2 → H_2+ H proceeds by bond exchange through the H_3 transition state, with proton transfer, ion formation and salt dissolution computed on the same footing.

5.4 Biomolecules

Nothing in the model changes at this scale; only the number of domains. One qualification must precede the table, because without it the entries would be read for more than they show. At this scale the runs are not blind. Two things are supplied: the *secondary structure*, i.e. which residues are helical, from which an $i \to i{+}4$ hydrogen-bond bias is generated; and a surface-area term with two constants standing in for the hydrophobic effect. The tertiary packing — how the helices arrange against one another — is *not* biased and is computed. Both supplied terms stand in for identifiable missing physics. Blind assembly fails because electron repulsion dominates at intermediate separations of three to eight Ångström while the attractive hydrogen bond reaches only about two, so nothing nucleates; and the hydrophobic effect originates in water entropy, to which a ground-state model of electron density has no access. What these entries therefore show is that the quantum model produces *correct hydrogen-bond geometry, and tertiary arrangement, at the scale of a protein* once the secondary structure is given — not that it predicts secondary structure or folds from sequence. The parameter-free hydrogen-bond results are the dimers and the ice of the previous section; these are the same physics carried to a larger system with the approach supplied.

system	result	reference
DNA G:C base pair	2 of 3 Watson–Crick bonds at ~ 2.1 Å	1.9 Å
RNA hairpin, 12 nt	all 4 stem pairs form, G1:C12 = 1.99 Å	2.0 Å
Trp-cage, 20 residues	4 of 5 helix H-bonds, $R_g = 7.2$ Å	near native
Villin HP35, 35 residues	3 helices, H-bonds at 2.0 Å, core F10–F17 = 6.1 Å	3-helix bundle
Myoglobin, 153 residues	helices A = 1.51, G = 2.15, H = 2.68 Å	partial fold
coiled coil, 2×15 residues	helix separation $12 \to 7.5$ Å, blind	water-driven docking

The RNA hairpin assembles its own secondary structure from a U-shaped start: stem paired, loop open, correct topology. The coiled-coil docking is blind — no bias toward the bound state — and is driven by the water.

5.5 Attribution: a discrepancy traced to the reduction

Accountability has a second half that is easy to overlook. A model is held to account not only when its predictions are checked but when its *failures* can be located — assigned to the model, or to a reduction of it, with the assignment itself open to test. The standard formalism cannot do this, as Section 4 argued: a discrepancy is charged to the functional or the basis because no computation of the equation exists to charge it to. One example shows RealQM in the opposite position.

Ionization energies come out badly in the radial solver. Beyond lithium the valence charge is too weakly bound, and by $Z \approx 14$ it is unbound altogether; the material was withdrawn from the companion article on atoms for that reason. The question is whether this indicts the model or the reduction, and the answer is decided by a bracket rather than by argument. The radial solver has no angular coordinate, so it cannot represent the octet as the model describes it — 4+4 at one radius, two dual tetrahedra angularly separated. It can represent only two alternatives, and for *every* noble gas from neon to radon the measured ionization energy falls between them: the octet split into two radial shells, with the outer tetrahedron too far out, gives about half the

measured value, while a single eight-shell, sitting too close to the kernel, gives two to three times it. The physical arrangement is geometrically intermediate between the two.

So the discrepancy is attributable, and attributable to something specific: a reduction that cannot express the model's own geometry, and which straddles the right answer because the two geometries available to it err in opposite directions. The attribution is also falsifiable, which is the point — a three-dimensional angular octet on the same pseudo-kernel either lands inside the bracket near the measured value or it does not. For neon the target is 21.6 eV against a bracket of 11.3–58. That calculation is listed among the tests in Section 9; bracketing shows the right answer is reachable, not that the model reaches it.

5.6 The scope is exercised, not asserted

This is the comparison that bears directly on universality, because it is about what each method can be *held to*:

method	largest system carried	hardware	time
Gaussian / VASP (DFT, HF) [11, 12]	**cannot fold proteins**	HPC cluster	days per point
GROMACS / OpenMM (MD)	20 residues	GPU clusters	weeks to month
Anton 2 (MD) [24]	80 residues	$100M custom ASIC	months per pro
RealQM	**153 residues, all electrons**	laptop GPU	minutes to hour

The quantum-chemical methods — the ones carrying the universality claim — appear in the first row, and they do not reach the problem. The two middle rows reach it with empirical force fields fitted to experiment and quantum calculation, so they are not tests of quantum mechanics at all. Machine-learning structure prediction [25] is a different kind of thing again: it predicts a structure from sequence without computing the physics, so it is not a test of any model of matter. RealQM carries the system from the charges and Z on commodity hardware — with, at this scale, the contact bias noted above. The claim here is about *scale*, which is what accountability requires: a model that cannot be evaluated on a protein cannot be checked against one. Whatever one concludes about accuracy, RealQM's scope is exercised rather than asserted, and that is the condition Section 1(3) asks for.

6 Scope and fidelity

One bound must be stated, because it marks the limit of what the record above establishes. RealQM is a fixed ansatz. Refining its mesh converges to the ansatz's own answer, not toward nature's, and there is no next order to add — no analogue of going from CCSD to CCSD(T) to full CI. So the claim supported here is universality of **scope**: the model applies across every scale of chemistry, is checked there, and agrees to a few per cent on energies and considerably better on geometry. It is not a claim of universal **fidelity**, which would require a convergence ladder that the model does not have. Being a testable claim, it is also refutable, and in places it has been refuted; that is the price and the point of being accountable. The components it does require are counted in Section 2, and they are fewer than the standard formalism's.

7 The asymmetry is dimensional

It would be easy to read the foregoing as a comparison of accuracy, and that is not the point. The two universality claims are of different *kinds*, and the dimension of the state is what makes them so.

A model whose state lives in $3N$ dimensions cannot be evaluated on a system of chemical interest, so its universality is a statement about an object that is never exhibited — an existence claim without a witness. A model whose state lives in 3 dimensions can be evaluated, so its universality is a prediction, and predictions can fail. This is the same distinction that separates a non-constructive existence proof from a computed solution, and it recurs across applied mathematics: the symbolic object is more general and cannot be inspected, the computed object is specific and can be checked. Neither is worthless, but they are not interchangeable, and universality means something different in each.

	standard QM	RealQM
state	$3N$ dimensions	3 dimensions
parameters	none, in the equation	none
cost	exponential in N	linear in N
species-blind	yes	yes
closed under composition	yes	yes
components beyond the energy	spin, antisymmetry, spin–statistics (3)	non-overlap (1)
evaluable on a protein	no	yes, 153 residues
verified range	1–3 electrons	H to Rn; H-bond geometry to 153 residues
verified precision	few parts in 10^5	few per cent; geometry better
exact in principle	yes	no
convergence ladder	yes, but unreachable	none
universality	**assumed**	**tested**

Standard quantum mechanics is verified at three electrons and asserted at 10^{23}. RealQM is verified from hydrogen to krypton on energies, and on geometry out to a 153-residue polypeptide. Over the range where either is actually used, the second is the better evidenced — a remark about evidence, not about truth.

8 What this establishes

It does not establish that standard quantum mechanics is false. Where it can be computed it is superb, and nothing here gives any reason to doubt it.

What it establishes is narrower and more useful. The conventional ranking treats the $3N$ formulation as the serious theory and a three-dimensional model as an approximation to be tolerated. On the specific question of universality that ranking inverts, because universality is a claim about scope and accountability, and only one of the two models is accountable over the scope it claims. A framework that applies in principle to everything and can be checked against nothing has a weaker title to universality than a model which applies to everything, is checked from the helium atom to a polypeptide, and agrees where it is checked. The model, its results and its code are set out at length elsewhere [26, 27, 28].

9 The tests that would decide

An argument that a model's universality is testable owes the reader the tests. Three are outstanding, each with a reference value not in dispute, and each capable of going against the model.

Neon's ionization energy. The attribution of Section 5.5 predicts that a three-dimensional angular octet — eight unit domains at cube-corner directions on a pseudo-kernel whose core radius comes from the radial solve — will land inside the bracket near the measured 21.6 eV, where the two radial arrangements give roughly 11 and 58. One number against an undisputed reference. If it falls outside the bracket, the attribution is wrong and the failure belongs to the model rather than to the reduction.

Dispersion, by upscaling rather than by addition. Two neutral closed-shell atoms attract as R^{-6}, and the model as it stands has no such term; nor does Hartree–Fock, which gives He_2 purely repulsive, while practical density functional theory supplies the term by fitting. Adding a fitted coefficient here would place RealQM in exactly the position Section 3 criticises, and should be refused. The legitimate route is upscaling. The dispersion coefficient follows from the dynamic polarisability by the Casimir–Polder integral,

$$\alpha(i\omega) = \sum_n \frac{f_n}{\omega_n^2 + \omega^2}, \qquad C_6 = \frac{3}{\pi}\int_0^\infty \alpha(i\omega)^2\, d\omega,$$

and the excitation energies ω_n and transition strengths f_n are quantities this model computes. So C_6 can be assembled from the model's own spectrum with nothing fitted and nothing added — a derived constant in the sense of Section 3, not an imported one. The construction carries its own audit: the sum must satisfy $\alpha(0) = \sum_n f_n/\omega_n^2$, which has to agree with the static polarisability computed directly. The test is then He_2, whose well is 0.035 mHa at $R = 5.6\, a_0$.

The static polarisability, which gates the other two. Polarisability is the one quantity here checked against spectroscopy rather than against another calculation, and it is also the audit on the dispersion construction. We do not have a converged value for helium: the available solvers disagree, and the number previously reported rested on a configuration carrying no electron–electron repulsion. Until that is settled nothing downstream of it can be trusted, which makes it the first of the three rather than the last.

What these have in common is worth stating, because it is the paper's thesis in miniature. Each is a calculation the model can be made to perform, against a number it did not choose, with an outcome that can embarrass it. That is what it means for a universality claim to be testable, and it is the one thing the claim examined in Section 4 cannot offer.

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